

1 Dinamical Systems

1.1 Models of Biological Systems

**Logistic Growth Model (Population):** Model:  $\frac{dN}{dt} = rN(1 - \frac{N}{K})$  N: population at time  $t$ ; K: carrying capacity (depends on resources);  $r, K > 0$ .

**Sol:**  $N(t) = \frac{N_0 K e^{rt}}{K + N_0(e^{rt} - 1)} \xrightarrow{t \rightarrow \infty} K$  **Equilibrium points:**  $N = 0$  (unstable),  $N = K$  (stable). **Quickest:**  $N = \frac{K}{2}$  ( $<$ : 凹;  $>$ : 凸)

**Delay Model:** Model:  $\frac{dN}{dt} = f(N(t), N(t-T))$  T: delay time  $> 0$  周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

<sup>1</sup>Ext. of Logistic Model:  $\frac{dN}{dt} = rN(t) [1 - \frac{N(t-T)}{K}]$  (用"ave. pop." 估计, 准用"Convolution Type")  $\Rightarrow$  <sup>2</sup>Dimensionless:  $\frac{dN^*}{dt^*} = N^*(t^*) [1 - N^*(t^* - T^*)]$  ( $N^* = \frac{N}{K}$ ,  $t^* = rt$ ,  $T^* = rT$ )  
<sup>1,2</sup> Periodic sol: stable limit cycle  $N(t + t_p) = N(t)$ ,  $t_p$ : period. **Amplitude:**  $rT$  **Equilibrium Points:**  $N = 0 | N^* = 0$  (unstable);  $N = K | N^* = 1$  (stable:  $0 < T^* = rT < \frac{\pi}{2}$ ; Unstable:  $T^* > \frac{\pi}{2}$ )

**Continuous Population Model (EN):** Model:  $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - EN = f(N)$   $r, K, E > 0$ ; EN: harvesting rate/unit time; E: effort measure.

**Equilibrium:**  $N = 0$  (unstable);  $N = N_h(E) = K(1 - \frac{E}{r}) > 0$  (need  $E < r$ , Stable) **Yield:**  $Y(E) = EN$   $\cdot E > r$ : die out; else: **Max Yield:**  $Y_M = \frac{rK}{4}$  ( $E = \frac{r}{2}$ ) **Steady State:**  $N_h | Y_M = \frac{K}{2}$

**Continuous Population Model ( $Y_0$ ):** Model:  $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$   $Y_0$ : constant yield rate/unit time.

**Max Yield:**  $Y_M = \frac{rK}{4}$  **Equilibrium:**  $0 < Y_0 < Y_M$ : Two positive points (unstable[小] and stable[大]). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

**Examples of  $N_{t+1} = f(N_t)$ :** Verhulst:  $N_{t+1} = rN_t(1 - \frac{N_t}{K})$ ;  $r, K > 0$ . Better: **Ricker curve:**  $N_{t+1} = N_t e^{r(1 - \frac{N_t}{K})}$

**Imperfect Bifurcation:**  $\dot{x} = rx - x^3 + h$  (codimension-2 bifurcation) If  $r \leq 0$ ,  $x^*$  unique. If  $r > 0$ ,  $|h| > h_c$ ,  $x^*$  unique;  $|h| < h_c$ ,  $x^*$  three. (两边 stable, 中 unstable). Bifurcation points:  $h = \pm h_c = \pm \frac{2r}{3} \sqrt{\frac{r}{3}}$ . **cusp point:**  $(r, h) = (0, 0)$

1.2 Summary - Stability and Bifurcation

|                                                        |                                                                                                                                                                                                |                                       |                                           |                                               |                                                                                                                                                   |  |                                                |  |                                                                                                                              |  |                                     |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------|
|                                                        | $\frac{dN}{dt} = f(N)$ No Limit cycle periodic solutions $S$ : stable; $U$ : unstable.                                                                                                         |                                       |                                           |                                               | $N_{t+1} = N_t F(N_t) = f(N_t)$ Eigenvalue: $\lambda = f'(N^*)$                                                                                   |  |                                                |  |                                                                                                                              |  |                                     |
| Stability                                              | Stable: $f'(N^*) < 0$                                                                                                                                                                          |                                       | Unstable: $f'(N^*) > 0$                   |                                               | Stable: $ \lambda  < 1$                                                                                                                           |  | Unstable: $ \lambda  > 1$ (chaotic)            |  | Periodic: $ \lambda  = 1 \begin{cases} + : \text{tangent bifurcation} \\ - : \text{period-doubling bifurcation} \end{cases}$ |  |                                     |
| Linearise                                              | $N(t) = N^* + n(t),  n(t)  \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} = \begin{cases} \text{(general)} = f(N^*) + f'(N^*)n(t) \\ \text{(delay)} = \text{直接带入 } t \text{ 后近似} \end{cases}$ |                                       |                                           |                                               | Let $N_t = N^* + n_t,  n_t  \ll 1 \Rightarrow N^* + n_{t+1} = \dots = f(N^* + n_t) \approx f(N^*) + f'(N^*)n_t \Rightarrow n_{t+1} = \lambda n_t$ |  |                                                |  |                                                                                                                              |  |                                     |
| Bifurcation \\<br>Perturbations<br>at $N^*$   Periodic | Saddle-Node (Normal Form)<br>$\dot{x} = r \pm x^2$                                                                                                                                             | Transcritical<br>$\dot{x} = rx - x^2$ | (super-)Pitchfork<br>$\dot{x} = rx - x^3$ | Sub-critical Pitchfork<br>$\dot{x} = r + x^3$ | $\lambda < -1$ : $\uparrow$ oscillations                                                                                                          |  | $-1 < \lambda < 0$ : $\downarrow$ oscillations |  | $0 < \lambda < 1$ : $\downarrow$ Monotone                                                                                    |  | $\lambda > 1$ : $\uparrow$ Monotone |
|                                                        |                                                                                                                                                                                                |                                       |                                           |                                               | If $f^p(N^*) = N^*$ , $N^*$ is $p$ -periodic. $\downarrow \forall$ even $p$ -periodic $N^*$ with $\lambda = -1$ branches into $2p$ -periodic      |  |                                                |  |                                                                                                                              |  |                                     |
|                                                        | $0 \rightarrow S \rightarrow S-S$                                                                                                                                                              |                                       |                                           |                                               | Sarkovskii's Theorem: If a 3-periodic $N^*$ exists, then $\forall p \geq 1$ , $p$ -periodic $N^*$ exists.                                         |  | Species extinction: $N_{\min} < 1$             |  |                                                                                                                              |  |                                     |
| Other                                                  | Time-Independent: At $N^*$ : $\frac{dN}{dt} = 0$ $N(t) = N(t-T) = N^*$ $f(N^*) = 0$                                                                                                            |                                       |                                           |                                               | Max/Min: (如果单峰 (unimodal)) 2-periodic $f'(N_m) = 0$ , $N_{\max} = f(N_m)$ ; $N_{\min} = f(N_{\max})$                                              |  |                                                |  |                                                                                                                              |  |                                     |
| Other                                                  | Tangency Condition: If $\dot{x} = f(x, r)$ , 分岔点 $(x^*, r_c)$ : $\frac{\partial f}{\partial x}(x^*, r_c) = 0$ $f(x^*, r_c) = 0$      两点合并: <b>coalesce</b> .                                   |                                       |                                           |                                               |                                                                                                                                                   |  |                                                |  |                                                                                                                              |  |                                     |

**Bifurcation:** Qualitative changes in dynamics are called *bifurcations*. (当平衡点性质 | 个数发生变化时) **Bifurcation Diagram:** y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

**稳定性 Stability 判别 | For Continuous Model:** Find perturbation:  $n(t)$  and  $\frac{dn}{dt} \Rightarrow$  Assume  $n(t) = e^{\lambda t} | ce^{\lambda t} \rightarrow$  Find  $\lambda = g(\lambda) \Rightarrow$  Let  $\lambda = u + iw \rightarrow u = g_1(u)$  and  $w = g_2(w) \Rightarrow u < 0$  asy.stable;  $u > 0$  unstable. **Find  $T_c$ :** (如要寻找参数 (e.g.  $T$ ) 范围/临界值 使得平衡点稳定) Check  $T = 0$  stable.  $\Rightarrow$  Using  $u = 0$  (i.e.  $\lambda = iw$ ) to find  $T_c$ .

2 Interacting Populations

**Lotka-Volterra Model (Predator-Prey):**  $\frac{dN}{dt} = N(a - bP)$ ;  $\frac{dP}{dt} = P(cN - d) \Rightarrow$  nondim:  $\frac{du}{d\tau} = u(1 - v)$ ;  $\frac{dv}{d\tau} = \alpha v(u - 1)$  **Equilibrium:**  $(u^*, v^*) = (0, 0), (1, 1)$

**Exact sol:**  $\alpha u + v - \ln(u^\alpha v) = H$ ,  $H > H_{min} = \alpha + 1$  constant. (closed phase) **Conservation Law:**  $\frac{dH}{d\tau} = 0$ , here it satisfies. (By chain rule)

**Show Level curves (constant of motion) closed:** If  $H(u, v)$  is level curve, Look  $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv} \end{bmatrix}_{(u^*, v^*)} \Rightarrow$  If  $+/-$  definite (i.e.  $\forall \lambda_i > 0 (< 0)$ )

$\Rightarrow$  It's local min/max at  $(u^*, v^*)$  & strictly convex/concave  $\Rightarrow$  closed curves. Closed curves  $\Rightarrow$  periodic sol.

**Determinant Stability:** For model:  $\begin{bmatrix} u' \\ v' \end{bmatrix} = \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix}$  at equilibrium  $(u^*, v^*)$ . Let  $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u - u^* \\ v - v^* \end{bmatrix}$ , then  $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} f_u & f_v \\ g_u & g_v \end{bmatrix}_{(u^*, v^*)} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow$  Find eigenvalues  $\lambda$  of  $J$  by  $\det(J - \lambda I) = 0$

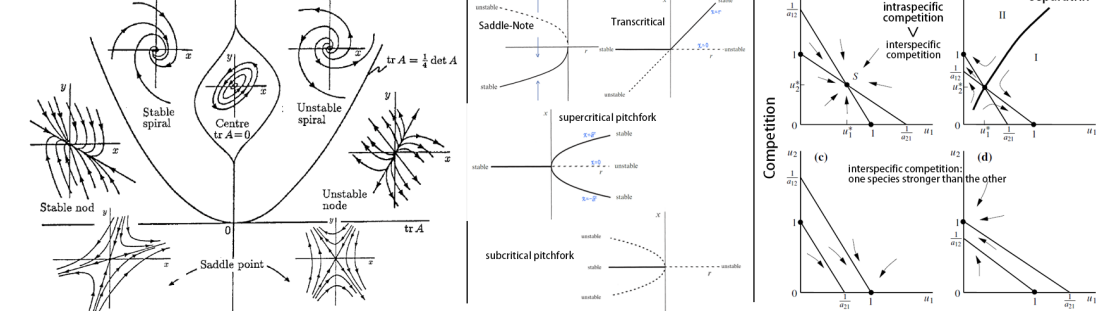
<sup>1</sup>  $\det(J) > 0$ ,  $\text{tr}(J) < 0 \Rightarrow \Re(\lambda) < 0 \Rightarrow$  stable node  $\text{Im}(\lambda) = 0$ ; stable focus=spiral point  $\text{Im}(\lambda) \neq 0$ ; <sup>2</sup>  $\det(J) > 0$ ,  $\text{tr}(J) > 0 \Rightarrow \Re(\lambda) > 0 \Rightarrow$  unstable node  $\text{Im}(\lambda) = 0$ ; unstable focus=spiral point  $\text{Im}(\lambda) \neq 0$

<sup>3</sup>  $\det(J) < 0 \Rightarrow \text{Im}(\lambda) = 0$ ,  $\lambda_1 < 0 < \lambda_2 \rightarrow$  saddle point (unstable); <sup>4</sup>  $\det(J) > 0$ ,  $\text{tr}(J) = 0 \Rightarrow \lambda = \pm iw$  (pure imaginary)  $\rightarrow$  center (neutrally stable)

**Null clines:**  $f(u, v) = 0$  or  $g(u, v) = 0 \rightarrow$  得  $u, v$  表达线条是 nullclines. **Trapping Region:** A closed region  $R$  可以用直线/null clines 构成 s.t. 所有 边界的外法线方向  $\mathbf{n}$  在对应边界上的点, 满足  $\mathbf{n} \cdot \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} < 0$

**Poincare-Bendixson Theorem:** One unstable node/focus/spiral point + it's in *trapping region*  $\Rightarrow$  A **stable limit cycle** exists. (Cannot be a saddle (since  $\det(J)$  cannot be  $< 0$ ))

**Competition Model: principal of competitive exclusion:** If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (b,c,d)



3 Reaction Kinetics

**Enzyme Kinetics:** e.g.  $S + E \xrightleftharpoons[k_{-1}]{k_1} SE \xrightarrow{k_2} P + E$   $K_i$ : rate of reaction;  $S$ : substrate;  $E$ : enzyme;  $SE$ : complex;  $P$ : product.

**Singular Perturbations Analysis:** For  $\frac{du}{d\tau} = f(u, v)$ ,  $\varepsilon \frac{dv}{d\tau} = g(u, v)$ ,  $0 < \varepsilon \ll 1 \Rightarrow v$  fast (when  $g \neq 0$ ),  $u$  slow. **Quasi-steady-state approximation:**  $\varepsilon \rightarrow 0 \Rightarrow g(u, v) = 0$

**Taylor:**  $u(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n u_n(\tau)$ ;  $v(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n v_n(\tau)$  **Outer solution:** Taylor 带入原方程组  $\rightarrow$  保留  $\mathcal{O}(1) | \varepsilon = 0$  Valid for  $\tau > 0$  (slow time);  $v_0(0)$  invalid.  $\rightarrow$  not uniformly valid for  $\tau \geq 0$

**Inner solution:**  $\sigma = \frac{\tau}{\varepsilon}$ ,  $\frac{U(\sigma; \varepsilon) = u(\tau; \varepsilon)}{V(\sigma; \varepsilon) = v(\tau; \varepsilon)} \Rightarrow \frac{dU}{d\sigma} = \varepsilon f(u, v)$ ,  $\frac{dV}{d\sigma} = g(u, v) \rightarrow$  Taylor 带入原方程组  $\rightarrow$  保留  $\mathcal{O}(1) | \varepsilon = 0$  (known as **boundary layer** ( $\tau = 0$  处)); Valid  $\tau = 0$  (fast time)

**Asymptotic Matching:**  $\lim_{\sigma \rightarrow \infty} [U_0(\sigma), V_0(\sigma)] = \lim_{\tau \rightarrow 0^+} [u_0(\tau), v_0(\tau)]$  **Uniformly Valid Solution:**  $u(\tau; \varepsilon) = u_0(\tau) + \mathcal{O}(\varepsilon)$ ;  $v(\tau; \varepsilon) = \begin{cases} v_0(\tau) + \mathcal{O}(\varepsilon), & 0 < \varepsilon \ll \tau \\ V_0(\sigma) + \mathcal{O}(\varepsilon), & 0 < \tau \ll \varepsilon \end{cases}$

**Activator-Inhibitor Model:**

4 Appendix

$y' + p(x)y = q(x)$ ; **Integrating Factor:**  $I(x) = e^{\int p(x)dx} \Rightarrow I(x)y = \int I(x)q(x)dx$  **Taylor Expansion:**  $f(x, r)$  at  $(x_0, r_0)$ :  $f(x, r) = f(x_0, r_0) + (x - x_0)f_x|_{(x_0, r_0)} + (r - r_0)f_r|_{(x_0, r_0)} + \frac{1}{2} \dots$

**Binomial Series:**  $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots$  **Linerisation:**  $f(x + \Delta x) = f(x) + f'(x)\Delta x + \dots$   $e^{bx} = 1 + bx + \frac{(bx)^2}{2!} + \dots$   $\ln(a + x) = \ln(a) + \frac{x}{a} - \frac{x^2}{2a^2} + \frac{x^3}{3a^3} - \dots$

**Hyperbolic Functions:**  $\sinh x = \frac{e^x - e^{-x}}{2}$   $\cosh x = \frac{e^x + e^{-x}}{2}$   $\tanh x = \frac{\sinh x}{\cosh x}$   $\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$   $\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$   $\tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$